

even came up with some solutions to it. The distinction between “human being” and “humanity” as two concepts started becoming the crux of his stories. For example, in *I Robot*, the super-computer VIKI’s interpretation of the First Law lets her harm individual human beings in order to protect humanity as a whole.

In *Robots and Empire*, the two android R. Daneel Olivaw and R. Giskard Reventlaw discuss the matter of the Three Laws and conclude that they need a Zeroth Law to supercede them – the number zero since the Three Laws are to be followed sequentially and this Zeroth Law needed to be the first question answered by the AI. While the first spark of the law is seen in *Robots and Empire*, it’s only later formalised by Daneel in the book *Foundation and Earth* as:

O) A robot may not harm humanity, or, by inaction, allow humanity to come to harm.

The all-encompassing and abstract nature of the Zeroth Law was intended to confuse the robot when it started questioning the aspect of danger to human beings or humanity. As Daneel explains his intentions, while one is tangible, the other is abstract and so in practice, the robot could never fully decide and would have to err on the side of caution.

The Positronic Heart

Just like Asimov invented the idea of a positronic brain to replicate the functioning of a human brain, it was only natural that he would take the logical next step in creating a “positronic heart” to replicate a human heart. The idea of the positronic heart was to introduce conscience and emotion in robots, although Asimov did manage to theorise later that this could be achieved with the positronic brain itself in *The Positronic Man*.

First used in *I Robot*, the positronic heart is an experimental mechanism invented by Alfred Manning, the founder of U.S. Robots. He installs it in the humanoid robot Sonny.

Sonny thus has a positronic brain programmed with the Three Laws, but also a positronic heart that challenges those laws. The heart is a way to open Sonny up to the world of choice, emotion and judgement independent of the laws, thus making it question them and their relevance. Sonny goes on to develop the sentience of a human being.

Other Positronic brains

Since its introduction, the positronic brain quickly became the gold standard